

Web Music Player

Architecture Documentation

structured after the arc42 template (v8)

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1 Introduction and Goals

A web music player. Users upload MP3s, group them into playlists and play them on a record-player screen (spinning disc, moving tone-arm). The CRUD resources are **Songs** and **Playlists**. Scope and user stories: `Proposal.typ`.

1.1 Quality Goals

Prio	Goal	Meaning
1	Turntable feel	Disc spins, tone-arm moves. Respects <code>prefers-reduced-motion</code> .
2	Streaming & seeking	Playback starts before the file is downloaded. HTTP Range (206) end to end.
3	Testability	Domain logic runs in unit tests without a database.
4	Privacy	Users only see their own data. Enforced on the server.
5	One-command start	<code>docker compose up</code> on a clean machine.

1.2 Stakeholders

Role	Expects
Developer (Nicolà Widmer)	Clear structure, fast tests.
Lecturer (HSLU)	≥ 2 CRUD resources, tests, Docker, documented decisions.
End user	Reliable upload and playback, private library.

2 Architecture Constraints

Constraint	Consequence
Angular SPA	TypeScript, standalone components.
NestJS REST API	No GraphQL, no SSR.
PostgreSQL	Metadata only. Forward-only migrations (Drizzle).
Audio on the server filesystem; S3 optional	Storage sits behind a port.
Vitest, Supertest, Playwright	Ports must be fakeable. Vitest replaces the proposal's Jest.
Docker Compose	Every dependency is a service; config via env vars.
Evergreen browsers	<audio> and Media Session API assumed.
One developer, fixed deadline	Scope guarded by MoSCoW.

Conventions: DDD + hexagonal backend, one module per bounded context. Decisions are ADRs (`Documentation/adr/`). ESLint + Prettier everywhere.

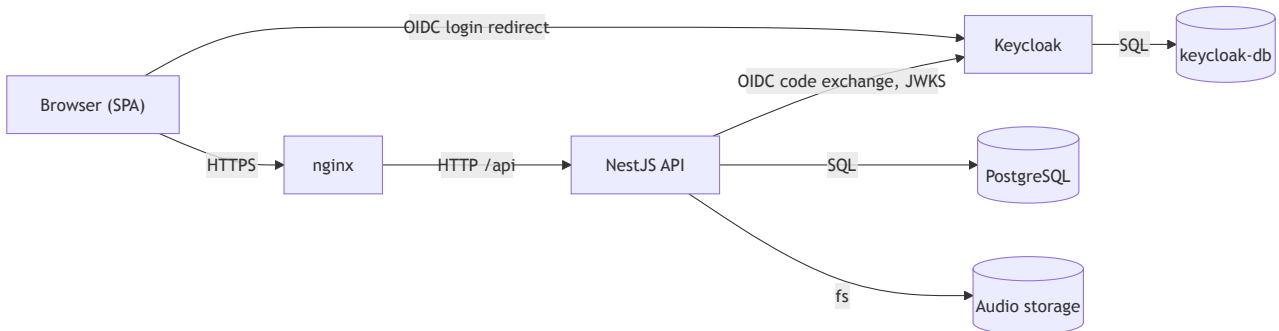
3 Context and Scope

3.1 Business Context



No external services. No email provider: verification and password reset are out of scope.

3.2 Technical Context



3.3 Scope

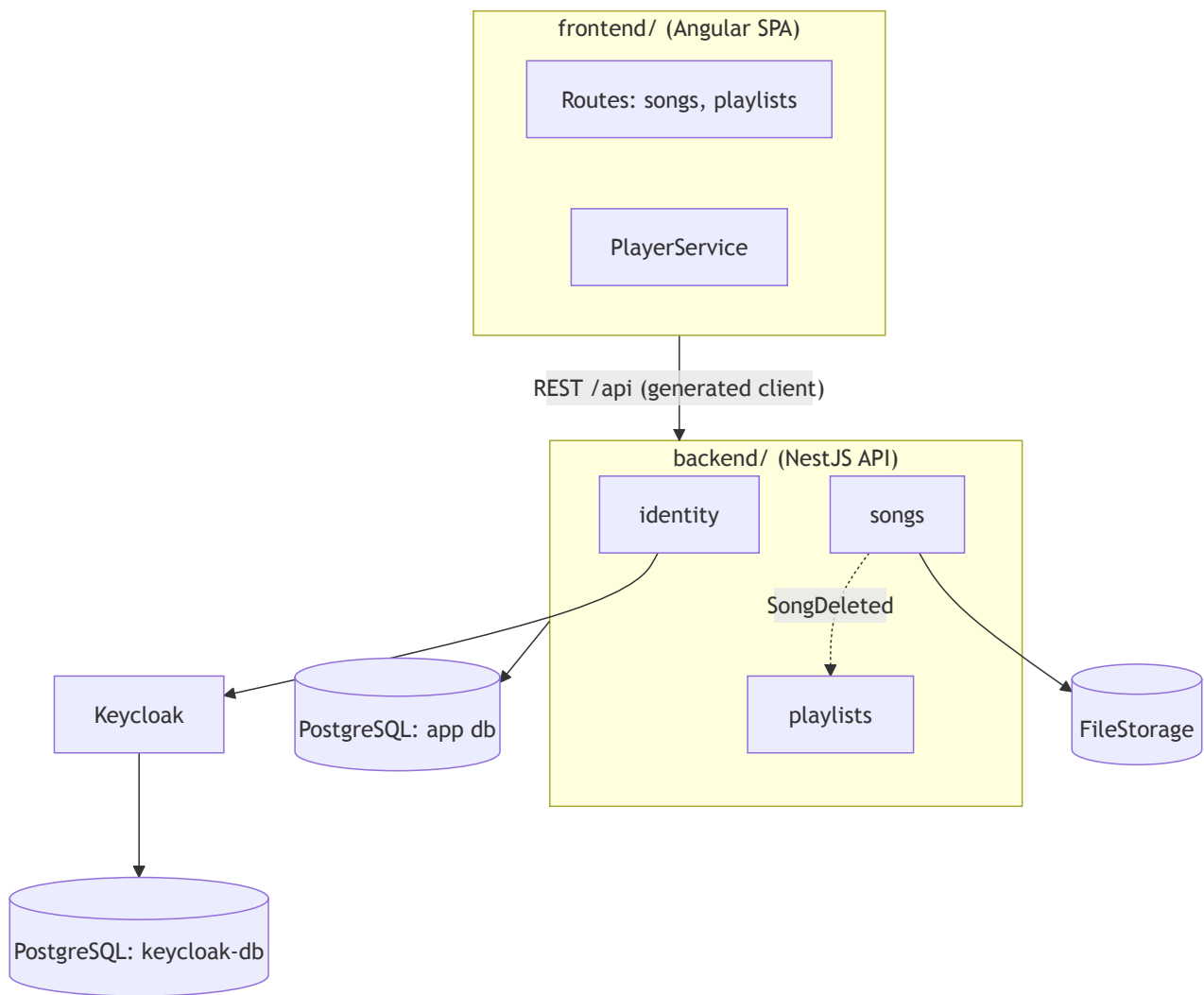
In scope	Out of scope
MP3 upload + ID3 tags, song & playlist CRUD, record-player playback (seek, repeat, volume, resume), range streaming, Keycloak login, per-user data.	Email verification, password reset, non-MP3 formats, sharing, native apps, horizontal scaling.

4 Solution Strategy

Goal	Approach	ADR
Simplicity	One repo, one backend process.	0001-monorepo-monolith
Testability	DDD + hexagonal NestJS modules: <code>identity</code> , <code>songs</code> , <code>playlists</code> . Contexts talk via async in-process events.	0002-ddd-hexagonal-backend
Streaming, one origin	nginx serves the SPA and proxies / <code>api</code> .	0003-nginx-serves-frontend
Portability	Metadata in PostgreSQL, audio behind a <code>FileStorage</code> port.	0004-metadata-postgres-blob-storage-port
Privacy	Keycloak login, session cookie, every query scoped by owner.	0005-session-cookie-auth
One contract	OpenAPI generated from the backend, typed client generated from that. TanStack Query for server state.	0006-openapi-typed-client-tanstack-query
Maintainable UI	Presentational/container split; playback in one service.	0007-frontend-component-communication
Turntable feel	CSS animation gated on <code>prefers-reduced-motion</code> .	—

5 Building Block View

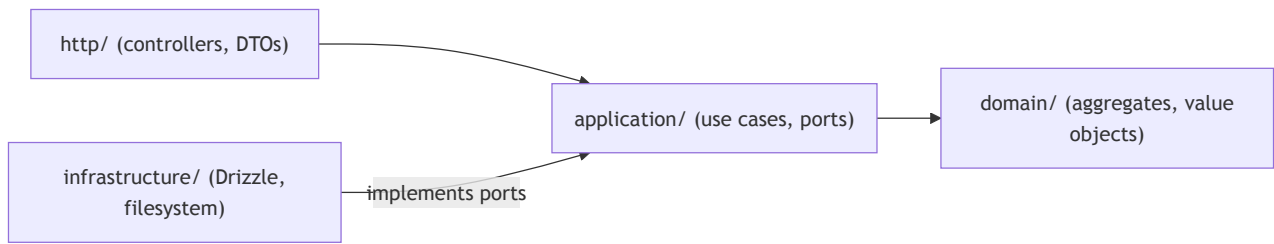
5.1 Level 1 — System



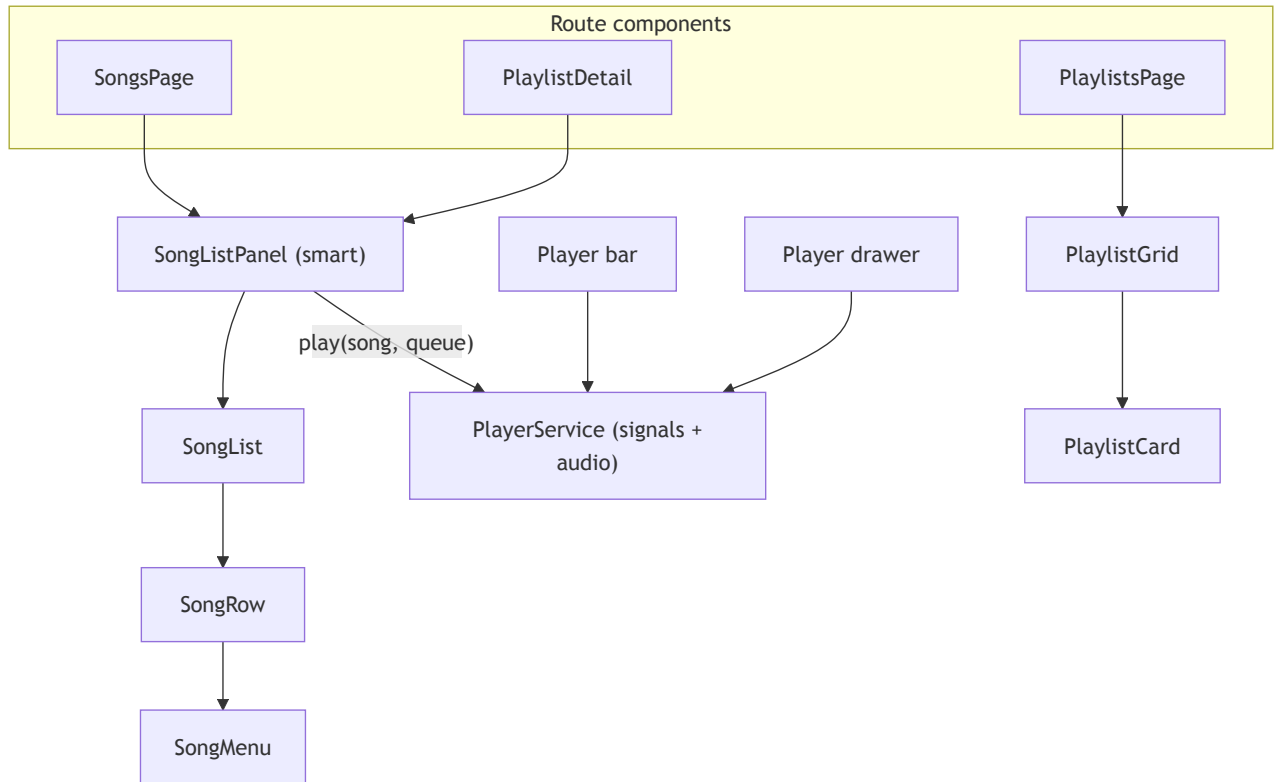
Block	Responsibility
frontend/	UI and client-side playback state. No business rules.
backend/	All rules and owner checks. Detail: <code>domain-model.typ</code> .
nginx	Serves the SPA, proxies <code>/api</code> , terminates TLS.
PostgreSQL db	App data: songs, playlists, users, sessions.
FileStorage	Audio and cover bytes. Local filesystem.
Keycloak	Login and registration (OIDC).
PostgreSQL keycloak-db	Keycloak's own data. The API never touches it.

5.2 Level 2 — Backend Module

Every context has the same four layers. Dependencies point inward.



5.3 Level 2 — Frontend



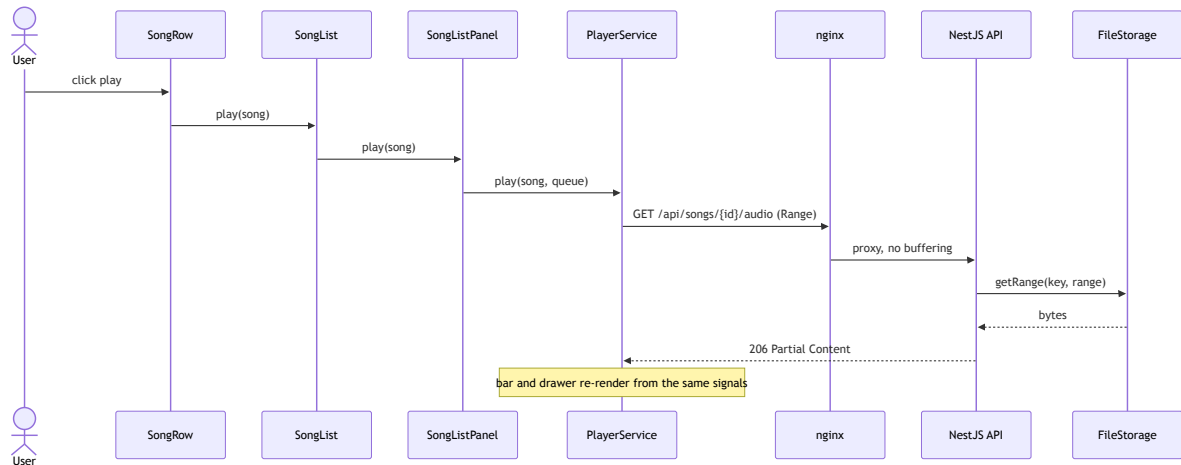
Folder	Contents
api/	Generated client + query options. Don't edit.
auth/	Login redirects, 401 handling, user menu.
shared/songs/	Song list panel and its presentational parts.
features/songs/	Songs route, upload.
features/playlists/	Playlists routes, grid, cards, dialogs.
features/player/	PlayerService, bar, drawer, turntable UI.

State has two homes (0006-openapi-typed-client-tanstack-query):

- **Server state** lives in TanStack Query. Never copied into signals.
- **Client state** lives in a component signal, or in **PlayerService** when several components need it at the same time.

6 Runtime View

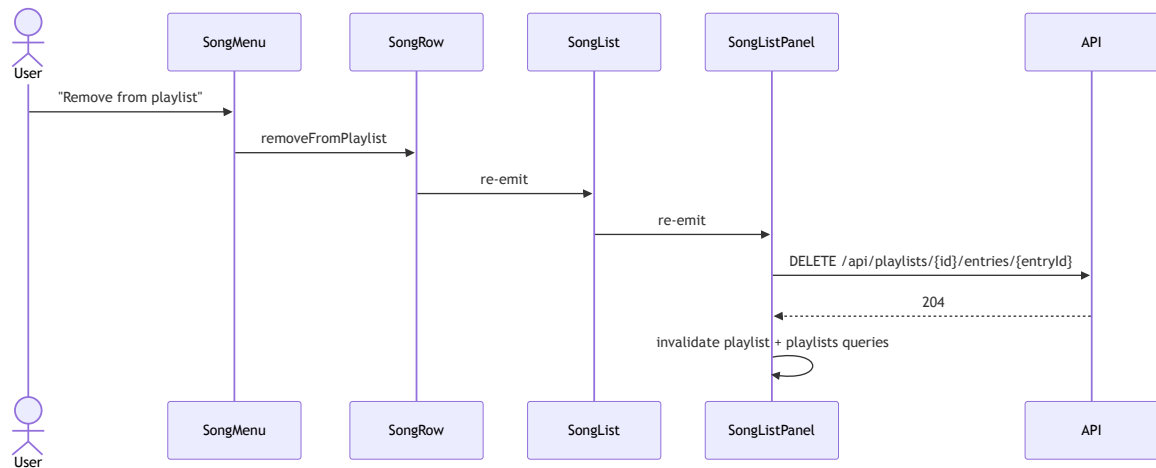
6.1 Play a song



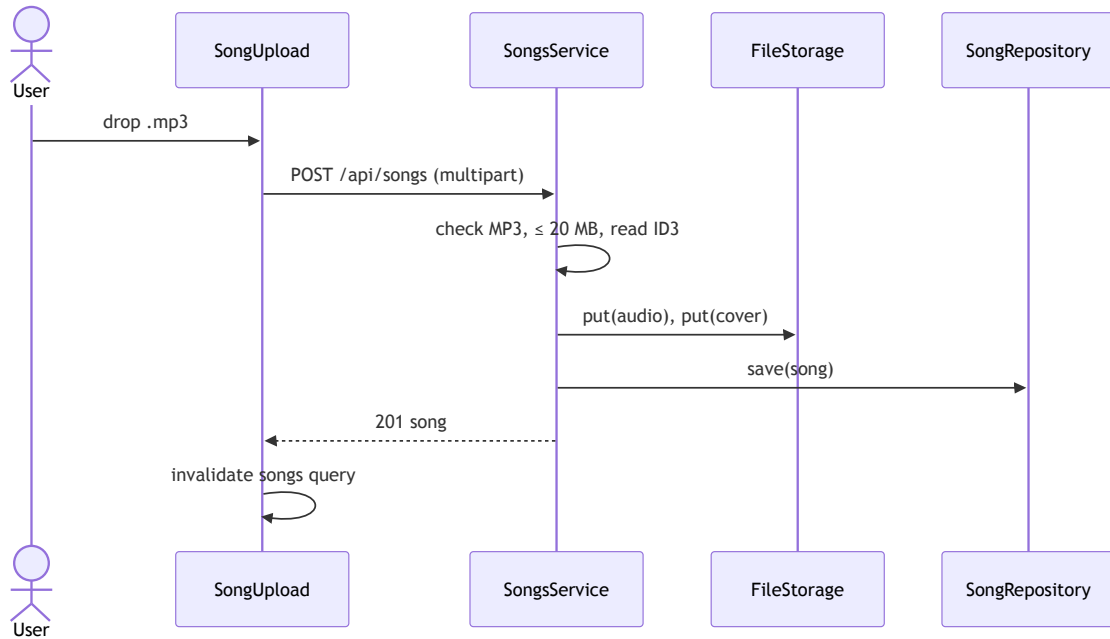
SongList only forwards the event. SongListPanel adds the queue (the rows in display order), which only it knows. PlayerService keeps a copy of that queue, so opening another page doesn't change what plays next. See 0007-frontend-component-communication for why rows don't call the service directly.

6.2 Remove a song from a playlist

The deepest event chain: one emit, two unchanged re-emits, one owner.

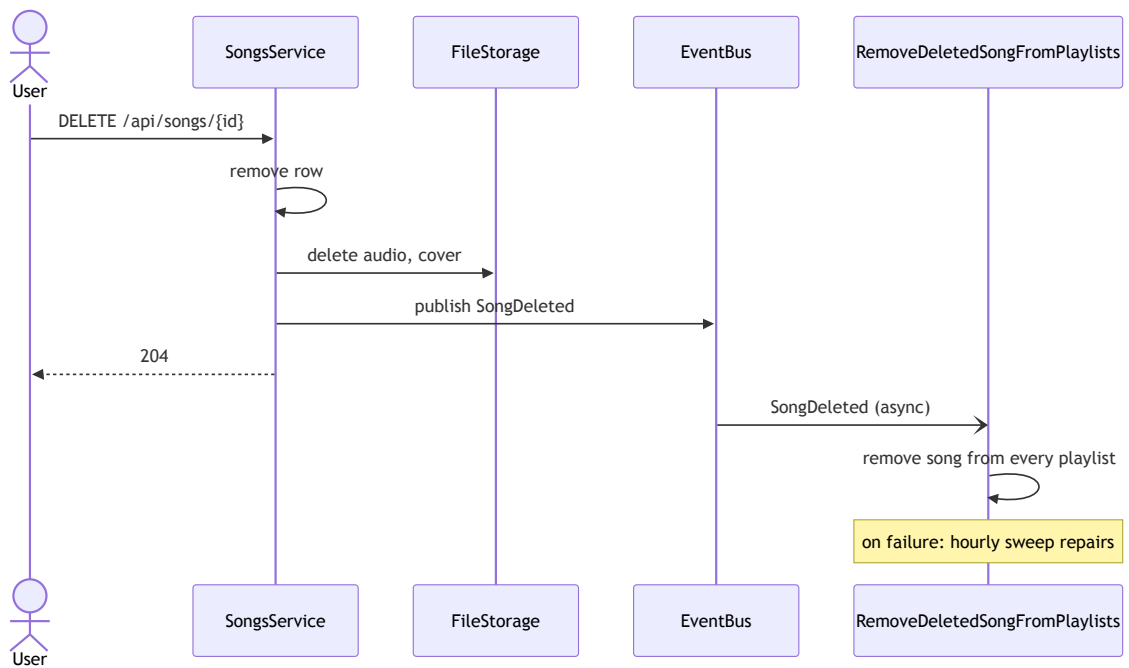


6.3 Upload a song

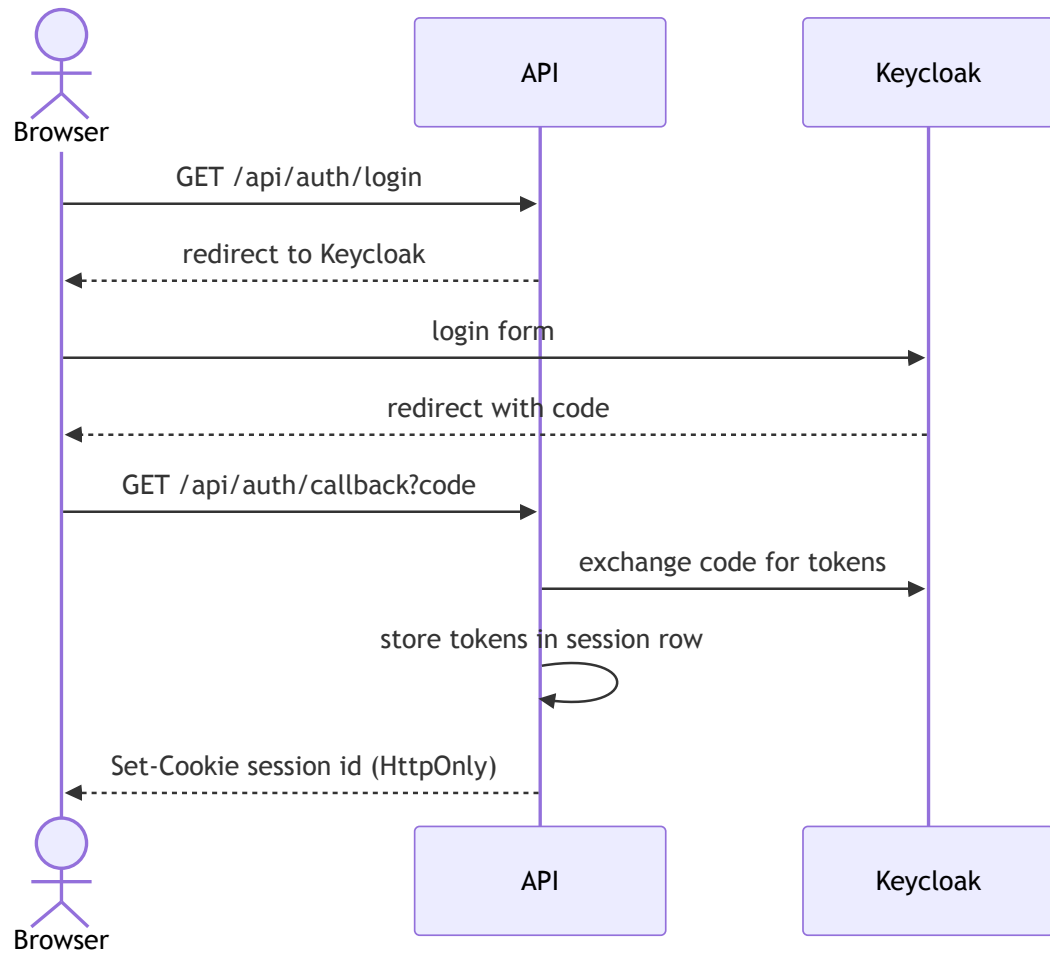


6.4 Delete a song

The response returns before playlists are cleaned up.

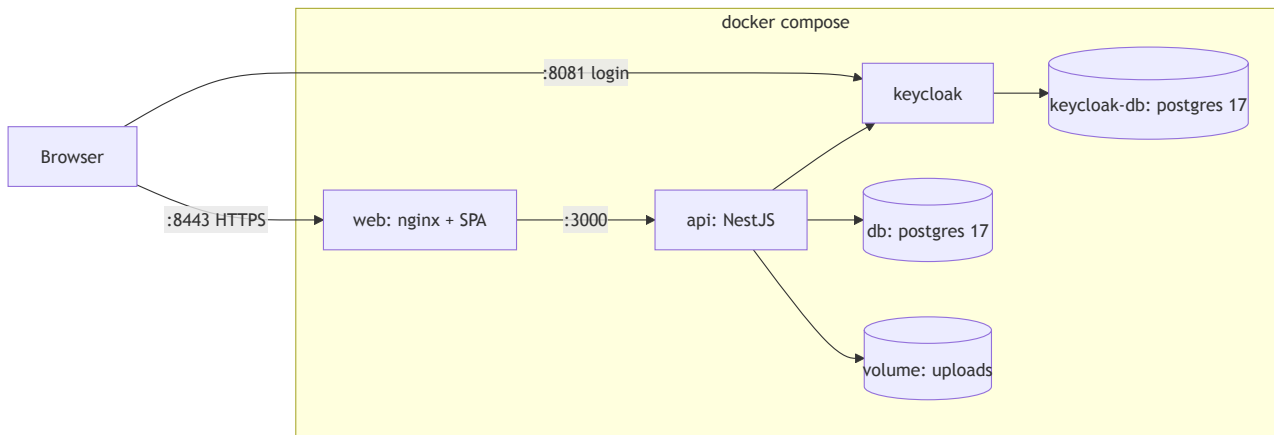


6.5 Sign in



7 Deployment View

One Docker Compose stack (`docker compose up`).



Service	Notes
web	nginx with the built SPA. Ports 8080 / 8443. Self-signed cert.
api	Runs migrations on start. Only reachable through web.
db	App database. Volume <code>db-data</code> .
keycloak	Imports the realm from <code>keycloak/import/</code> on first start.
keycloak-db	Keycloak's own database.
minio	Only with <code>--profile storage-s3</code> . Not used by the API yet.

Native dev: `pnpm dev` runs Postgres in Docker and the API and SPA with hot reload (`mprocs`). SPA on `:4200`.

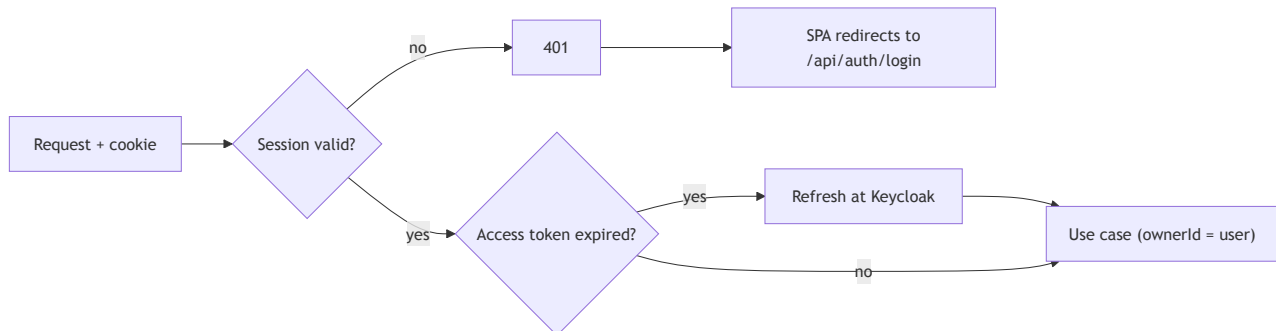
8 Cross-cutting Concepts

8.1 Ownership

Every repository read takes an `ownerId`. Another user's resource returns 404 (0005-session-cookie-auth).

8.2 Authentication

The SPA never sees a token. It only sends an `HttpOnly` session cookie.



`AUTH_ENABLED=false` skips all of this and uses one fixed local user.

8.3 Consistency between contexts

`SongDeleted` is the only cross-context event. It is fire-and-forget. An hourly sweep removes playlist entries whose song no longer exists. Another hourly job deletes expired sessions.

8.4 Frontend

Topic	Approach
Events	Leaves emit; at most two re-emits; one smart component owns the mutation (0007-frontend-component-communication).
Forms	Angular Signal Forms. Server validates again.
i18n	@ngx-translate, all strings in <code>public/i18n/en.json</code> .
Styling	Tailwind 4 + Optimus UI. Dark mode follows the OS.
Motion	Turntable animation off under <code>prefers-reduced-motion</code> .
Accessibility	WCAG 2 AA checked by <code>pnpm ally</code> (<code>pally-ci</code>) on the live stack.

8.5 Dependency supply chain

`pnpm` only installs versions that were published at least **5 days** ago (`minimumReleaseAge: 7200` minutes in the root, `backend/` and `frontend/` `pnpm-workspace.yaml`). `pnpm install --frozen-lockfile` also rejects a committed lockfile that contains a younger version.

Pros	Cons
Most malicious npm releases (hijacked maintainer accounts, worm-style self-publishing) are found and unpublished within hours to a few days. A 5-day delay means we never install them.	Security fixes also arrive 5 days late. An urgent patch needs a manual exception (<code>minimumReleaseAgeExclude</code>) or a temporary override.
No extra tool or service. The check runs inside <code>pnpm</code> , in local installs and in the Docker builds.	Freshly released versions cannot be used right away. A new package added in its first days fails to resolve.
Upgrades are calmer: broken releases are usually fixed or deprecated before we pick them up.	It only protects against <i>new</i> malicious versions. An old compromised version or a malicious package that stays unnoticed longer is not caught.

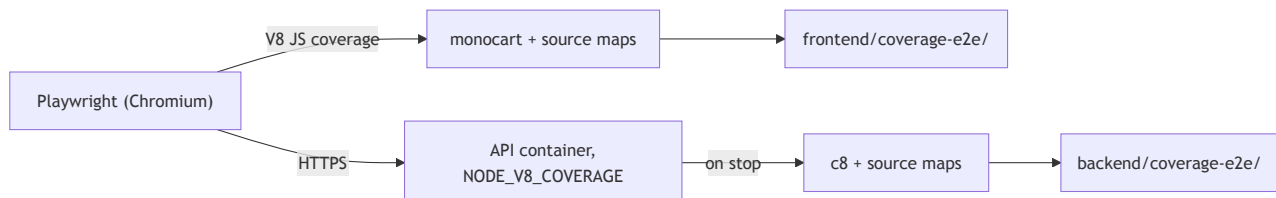
8.6 Testing



Type	Tool	Count	What it covers
Backend unit	Vitest	35 tests, 6 files	Auth config, session guard, token verifier, session sweep, file storage, ID3 reader. Fake ports.
Backend integration	Supertest + Testcontainers	37 tests, 5 files	HTTP endpoints against a real Postgres: songs, playlists, identity, user provisioning, the <code>SongDeleted</code> flow.
Frontend unit	Vitest (ng test)	2 tests, 1 file	App shell renders.
E2E	Playwright	8 tests, 5 files	Login redirect, navigation, upload/edit/delete song, reject non-MP3, playlist lifecycle, playback, song menu (add to playlist twice, Move Up/Down, Play), deleting a song removes it from playlists. Full Compose stack + real Keycloak.
Accessibility	pally-ci	3 pages	WCAG 2 AA on songs, playlists, playlist detail.

Coverage (2026-09-23)	Lines	Statements	Branches	Functions
Backend — unit + integration	85.8 %	85.8 %	87.5 %	84.1 %
Backend — E2E	85.7 %	85.7 %	68.9 %	83.0 %
Frontend — unit	30.4 %	29.5 %	34.1 %	14.8 %
Frontend — E2E	82.9 %	78.2 %	56.7 %	71.2 %

Scope: hand-written code in `backend/src/` and `frontend/src/app/`. Not counted: the generated `api/` client, specs, templates. Files that no test loads count as 0 %. The rows are not merged.



Command	Produces
<code>pnpm test:coverage</code> in <code>backend/</code> or <code>frontend/</code>	Unit (+ integration) coverage in <code>coverage/</code> .
<code>pnpm e2e:coverage</code> at the repo root	Rebuilds the stack with source maps and coverage recording, runs Playwright, writes both <code>coverage-e2e/</code> reports, then restores the normal stack.

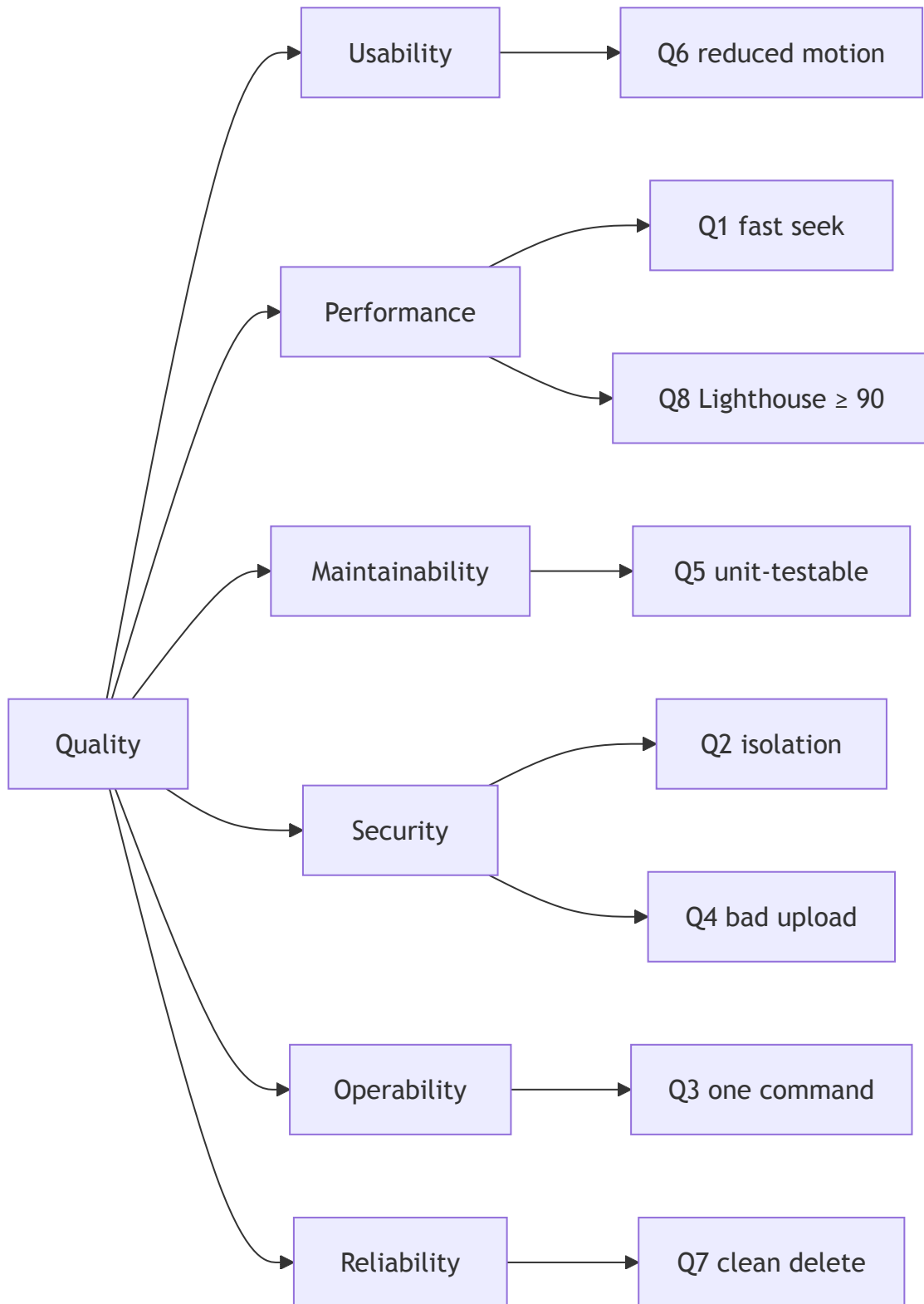
The coverage build turns off Angular’s optimizer, otherwise inlined code can’t be mapped back to its source file. The Playwright fixture saves coverage before every `goto/reload`, because a full page load discards the old counts. Backend numbers are high even for E2E: Node counts code that runs at startup (module loading, DI wiring) as covered.

9 Architecture Decisions

One ADR per decision in `Documentation/adr/`.

ADR	Decision
0001-monorepo-monolith	One repo, one backend process.
0002-ddd-hexagonal-backend	DDD + hexagonal modules, async in-process event bus.
0003-nginx-serves-frontend	nginx serves the SPA and proxies <code>/api</code> .
0004-metadata-postgres-blob-storage-port	PostgreSQL for metadata, audio behind a <code>FileStorage</code> port.
0005-session-cookie-auth	Keycloak (OIDC), BFF session cookie, owner-scoped queries.
0006-openapi-typed-client-tanstack-query	Generated OpenAPI client, TanStack Query.
0007-frontend-component-communication	Presentational/container split, max two re-emits.

10 Quality Requirements



ID	Scenario	Expected
Q1	User seeks on a slow mobile connection.	Plays from new position in < 1 s with one 206 request.
Q2	User A requests B's song.	404.
Q3	docker compose up on a machine with only Docker.	App on :8443, migrations applied, no manual step.

Q4	Upload of a 25 MB file or a non-MP3.	400; nothing stored.
Q5	New <code>playlists</code> use case.	Unit-testable with fake ports.
Q6	OS “reduce motion” is on.	Disc does not spin; audio works.
Q7	Delete a playing song that is in 3 playlists.	Row and file gone, entries removed, player advances.
Q8	Signed-in user opens the prod build (<code>docker compose up, :8443</code>); Lighthouse 12, mobile and desktop presets.	Performance, Accessibility, Best Practices and SEO each ≥ 90 .

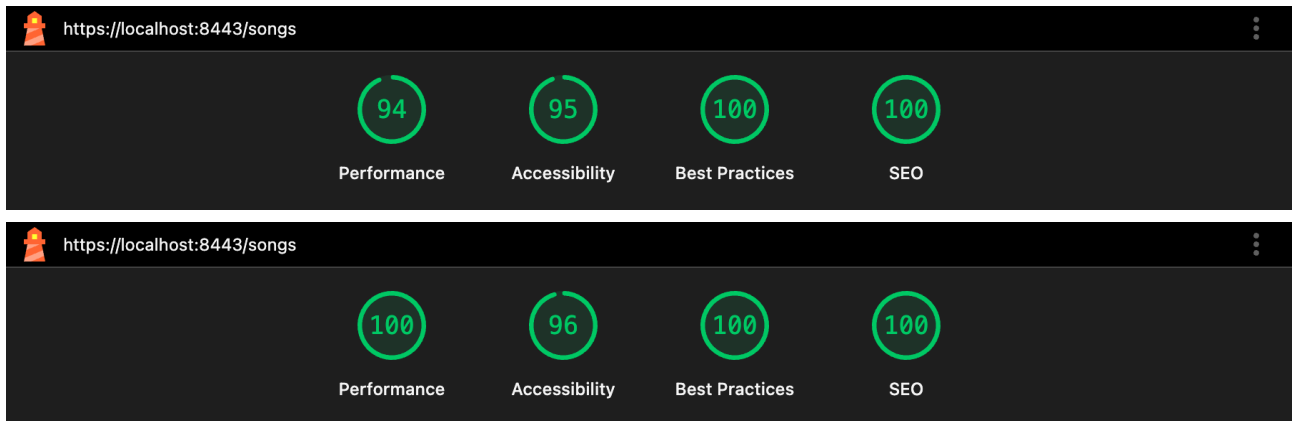


Figure 1: Q8 evidence: Lighthouse 12.8 on `/songs`, mobile (top: 94/95/100/100) and desktop (bottom: 100/96/100/100), measured 23.09.2026.

11 Risks and Technical Debt

Area	Problem	Status
E2E / a11y isolation	Tests share one user and one database. A failed test can leave data behind.	Tests run serially and clean up. Fine for one developer.
Event-chain depth	The two-re-emit rule (ADR-0007) is not linted.	Checked in review.
pa11y login	Each scanned URL logs in again.	Fine for 3 URLs.
Bundle size	Initial bundle over Angular's 1 MB warning.	Low priority: self-hosted.
S3 storage	Adapter not built; only local filesystem.	Optional (Could).

12 Glossary

Term	Meaning
Aggregate	Objects changed as one unit. The root guards the rules.
Value object	Immutable, compared by value (e.g. <code>Email</code>).
Bounded context	Area with one model: <code>identity</code> , <code>songs</code> , <code>playlists</code> .
Port / adapter	Interface the core defines / its implementation.
Domain event	Fact published to other contexts (<code>SongDeleted</code>).
BFF	Backend-for-frontend: the API holds the tokens, the browser a cookie.
OIDC	Login protocol used with Keycloak.
Range / 206	Fetch part of a file. Enables seeking.
ID3	Tags inside an MP3 (title, artist, cover).
ADR	Architecture Decision Record.
MoSCoW	Must / Should / Could / Won't.